

1. Introduction

Protein-protein interactions (PPIs) are fundamental to understanding cellular functions and biological processes. Our project focuses on analyzing PPI data to extract meaningful features that can be used for downstream computational modeling, such as predicting interactions between proteins. The aim is to create a reliable dataset and extract relevant protein features for machine learning applications.

2. Dataset Collection

2.1 Sources

We have collected protein sequences from **UniProt** databases for two organisms:

- **E. coli** – sequences up to 300 amino acids.
- **Human** – sequences up to 300 amino acids.

The FASTA files for these datasets were uploaded and stored for further analysis:

- `ecoli_uniprot_300aa.fasta`
- `human_uniprot_300aa.fasta`

2.2 STRING Database Mapping

We mapped protein IDs from the **STRING** database to UniProt IDs to integrate interaction information with protein sequence features. This step ensures that each protein in our dataset has corresponding PPI information.

3. Data Preprocessing

3.1 Redundancy Removal

- Exact duplicate sequences were removed to reduce bias and redundancy.
- Similarity thresholds were considered, but for now, only exact duplicates were eliminated.

3.2 Data Cleaning

- Removed incomplete or corrupted sequences.

- Checked for proper FASTA formatting and ensured consistency between sequence IDs and headers.
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4. Feature Extraction

We aim to extract **protein-level features** for machine learning models. So far, we have identified several categories:

4.1 Sequence-Based Features

- **Amino acid composition** – frequency of each amino acid in the protein.
- **Dipeptide composition** – frequency of consecutive amino acid pairs.
- **Protein length** – number of amino acids.

4.2 Structural Features (planned)

- Predicted secondary structure content (alpha-helix, beta-sheet, coil).
- Solvent accessibility or exposure of residues.

4.3 Physicochemical Properties

- Hydrophobicity, polarity, charge, and molecular weight.
- These properties are relevant for protein binding and interaction propensity.

4.4 Rationale for Feature Selection

- Sequence composition features help capture the protein's identity.
 - Physicochemical properties provide insight into interaction potential.
 - Structural features indicate folding and potential binding sites.
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5. Tools and Implementation

- Python scripts are being used to parse FASTA files and compute features.
 - Sorting algorithms (like bubble sort and selection sort) were discussed for organizing data, though Python libraries like pandas may be used for efficiency.
 - Folder structure is maintained for dataset, feature scripts, and processed results for reproducibility.
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6. Current Progress

- Datasets collected and cleaned.
 - Redundant sequences removed.
 - Feature categories finalized for extraction.
 - Scripts prepared for amino acid composition and other sequence-based features.
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7. Next Steps

1. Extract all sequence-based features for both E. coli and human datasets.
 2. Compute physicochemical properties for all proteins.
 3. Incorporate structural predictions.
 4. Integrate extracted features with PPI information for machine learning models.
 5. Perform exploratory data analysis to identify patterns in interactions.
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8. Conclusion

We have successfully gathered and cleaned the datasets and finalized the feature extraction plan. This structured approach ensures high-quality data and meaningful features for further PPI prediction and analysis.